



**Generation Interconnection
Impact Study Report
for
Queue Project AD2-035
ONTELAUNEE 230 KV
14 MW Capacity / 38.9 MW Energy**

June 2019

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Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 36.2, as well as the System Impact Study Agreement between **Ontelaunee Power Operating Company, LLC**, the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Mid-Atlantic Interstate Transmission, LLC (MAIT).

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

General

The Interconnection Customer (IC) has proposed an uprate to an existing natural gas generating facility located at 5115 Pottsville Pike, Reading, Pennsylvania. This projects requests an increase to the install capability of **38.9** of uprate MW with **14** of uprate MW of this output being recognized by PJM as Capacity. The installed facilities will have a total capability of **630 MW** with **555.1 MW** of this output being recognized by PJM as Capacity. The proposed in-service date for this project is **December 1, 2019**. **This study does not imply a TO commitment to this in-service date.**

Queue No.	Maximum Facility Output
A18	557 MW
AA1-043	34.1 MW
AD2-035	38.9 MW
Total	630 MW

Queue No.	Capacity
A18	521.5 MW
AA1-043	19.6 MW
AD2-035	14 MW
Total	555.1 MW

Queue Number	AD2-035
Project Name	ONTELAUNEE 230 KV
Interconnection Customer	Ontelaunee Power Operating Company, LLC
State	Pennsylvania
County	Berks
Transmission Owner	ME (MAIT)
MFO	630
MWE	38.9
MWC	14
Fuel	Natural Gas
Basecase Study Year	2022

Point of Interconnection

AD2-035 “Ontelaunee 230 kV” uprate project will be an increase to the existing Ontelaunee Power Generating Plant which currently interconnects with the ME transmission system at the Ontelaunee Substation 230 kV bus. See Figure1 for the Project Location and Figure 2 for the One Line Diagram.

Cost Summary

The AD2-035 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$0
Direct Connection Network Upgrade	\$0
Non Direct Connection Network Upgrades	\$0
Total Costs	\$0

In addition, the AD2-035 project may be responsible for a contribution to the following costs

Description	Total Cost
System Upgrades	\$0

Transmission Owner Scope of Work

Attachment Facilities

There is no Attachment Facilities work associated with this project.

Direct Connection Cost Estimate

There is no Direct Connection work associated with this project.

Non-Direct Connection Cost Estimate

There is no Non-Direct Connection work associated with this project.

Incremental Capacity Transfer Rights (ICTRs)

No network impacts were identified so no study is required for an increase to the CETL in the 2021/2022 BRA case.

Schedule

No work is required by the TO to accommodate this interconnection.

Transmission Owner Analysis

Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2021 summer peak load flow model and the results were verified by FE. Additionally, FE performed an analysis of its underlying transmission <100 kV system. The AD2-035 project did not contribute to any overloads on the FE transmission system.

Short Circuit Analysis

Because the machine parameters will not change as a result of this project, a short circuit analysis is not required.

Stability Analysis

A dynamic stability analysis was completed by PJM and the results were reviewed by FE. The results of the stability analysis will be provided to the IC in PJM's System Impact Study report. There were no stability concerns identified.

System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the IC is responsible for the following:

1. The purchase and installation of the minimum required FE generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
2. The purchase and installation of supervisory control and data acquisition ("SCADA") equipment to provide information in a compatible format to the FE Transmission System Control Center.
3. Compliance with the FE and PJM generator power factor and voltage control requirements.
4. The execution of a back-up service agreement to serve the customer load supplied from the AD2-035 generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.

The IC will also be required to meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FE system.

Power Factor Requirements

The existing 557 MW portion of the Customer Facility shall retain its ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.90 lagging (supplying VARs) measured at the generator's terminals. The increase of 34.1 MW to the Customer Facilities associated with the AA1-043 project shall be designed with the ability to maintain a Power Factor of at least 1.0 (unity) to 0.90 lagging (supplying VARs) measured at the generator's terminals. The increase of 38.9 MW to the Customer Facilities associated with the AD2-035 project shall be designed with the ability to maintain a Power Factor of at least 1.0 (unity) to 0.90 lagging (supplying VARs) measured at the generator's terminals.

FE performed a reactive capability analysis of the proposed incremental increase. The following tables show the reactive power requirements and capabilities for both the existing and additional output.

Combustion Turbines

Queue	Size	Leading PF	Lagging PF	Leading MVAR	Lagging MVAR
A18	187.0	0.95	0.90	61.5	90.6
AA1-043	3.9	1.00	0.90	0.0	1.9
AD2-035	12.6	1.00	0.90	0.0	6.1
Total	203.5	-	-	61.5	98.6
Reactive Capability at 203.5 MW				97.5	138.5
MVAR Surplus/Deficiency				+36.0	+39.9

Steam Turbine

Queue	Size	Leading PF	Lagging PF	Leading MVAR	Lagging MVAR
A18	183.0	0.95	0.90	60.1	88.6
AA1-043	26.3	1.00	0.90	0.0	12.7
AD2-035	13.7	1.00	0.90	0.0	6.6
Total	223.0	-	-	60.1	108.0
Reactive Capability at 223.0 MW				45.0	57.5
MVAR Surplus/Deficiency				-15.1	-50.5

As shown in the tables above, both combustion turbines have an excess reactive power capability while the steam turbine has a reactive power deficiency. However, because the steam turbine cannot operate without one or both of the combustion turbines, at full operation with both combustion turbines and the steam turbine online, the plant will have an overall reactive power surplus of 56.9 leading MVAR and 29.3 lagging MVAR. If only one combustion turbine is online with the steam turbine, it is expected that the power output of the steam unit will be reduced. At the lower output of the steam turbine, the machine will have greater reactive power capability and will not be deficient.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

ME Requirements

The IC will be required to comply with all FE revenue metering requirements for generation interconnection customers which can be found in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>

Network Impacts

The Queue Project AD2-035 was evaluated as a 38.9 MW (Capacity 14 MW) injection at Ontelaunee 230kV in the ME area. Project AD2-035 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD2-035 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Load Flow

Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

System Reinforcements

None.

Flow Gate Details

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gage other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

None.

Affected Systems

LG&E

None.

MISO

None.

TVA

None.

Duke Energy Progress

None.

NYISO

None.

Short Circuit

Short Circuit

The following Breakers are overduty:

None.

Stability Analysis

Stability

No mitigations were found to be required.

Project Location

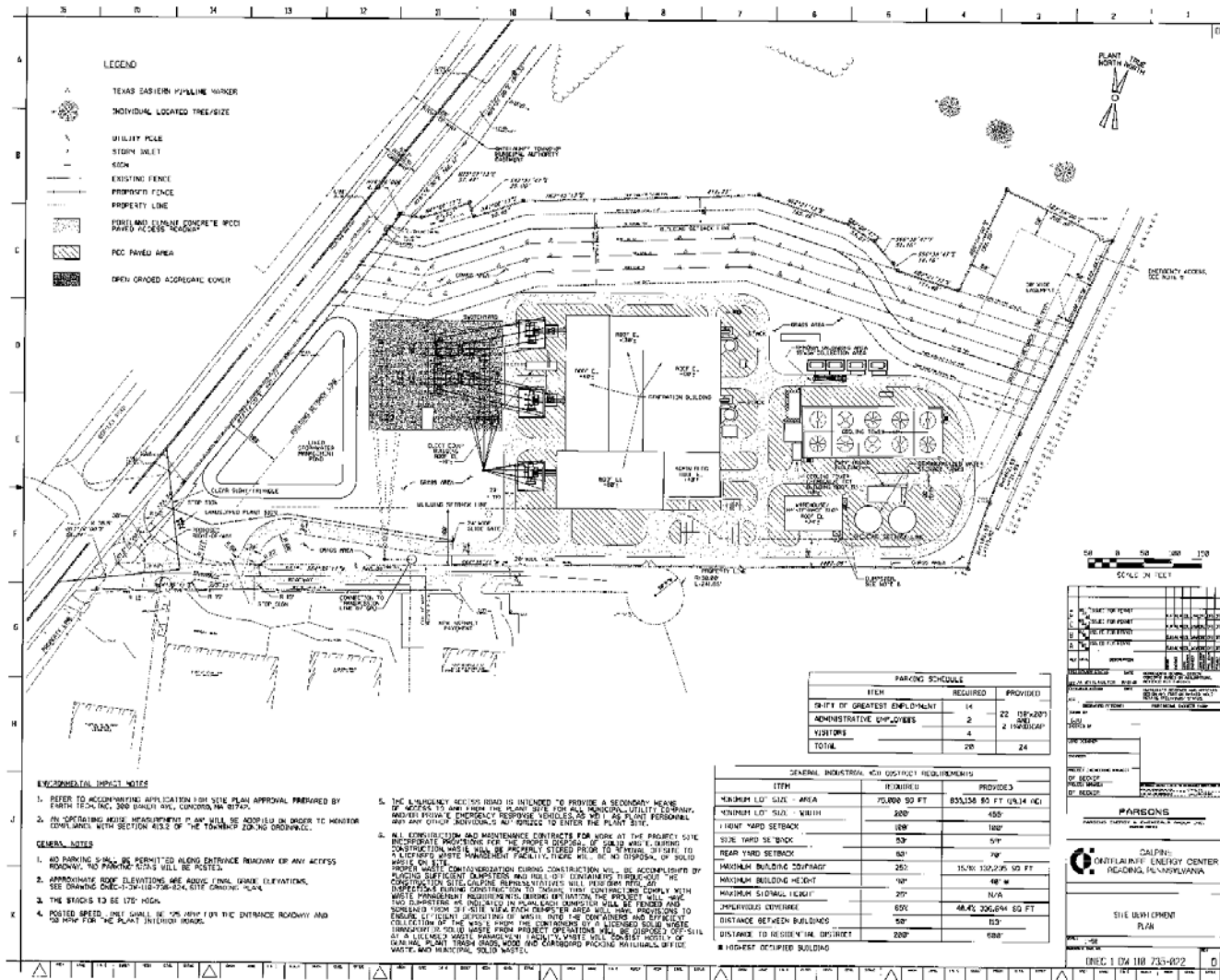


Figure 2

AD2-035 One Line Diagram

